A

Was Hamilton's 1790 debt assumption fair?

Reconciliation of $21.5M of assumed state war debt through five Treasury ledgers, 1790-1793. Sources: Founders Online (National Archives), U.S. Treasury, First Census. github.com/josephrbarry/hamilton-debt-assumption

Prepared by R. Barry
1790 $ · [2025 $ at 36.3x]

Authorized, Funding Act 1790	Assumed by 1793	Assumed, in 2025 $	Quota unused	Utilization, %	South Carolina quota per head (mean $5.31)
$21.5M	$18.27M	$663M	$3.23M	85	$16.06

Opinion

I believe Hamilton's allocation is defensible, based on the data in this report. The question isn't whether the allocation itself was tidy; it was whether the new federal government could take a $21.5 million problem off thirteen balance sheets and make it one credible obligation. It did. 85% of the authorized sum was taken up; two of the three states that came in over quota (MA, SC) were the two Hamilton's schedule ranked as owing the most; and within three years the debt was funded. Despite the 1792 panic, U.S. bonds traded near or above par.

The method was rough, and the records show it. Six of thirteen states filed returns; four got quotas by round number; MA and SC were cut by a quarter. But Hamilton was allocating with the data that existed in January 1790, and he said so in his report to Congress.

The 1793 settlement vindicated him. The audit Madison demanded found the biggest recipients (MA, SC) were also the biggest creditors of the Union. Madison's own Virginia turned out to owe the Union $100,879 - 12 cents a head, break-even as the Compromise intended - while receiving $2.9M of relief.

Where I'd push back: the claim that federal bonds would bind creditors to the new government can't be tested at the state level, and I don't assume that. What the numbers do show is a fair outcome, reconciled in about three and a half years (Jan 1790 report to Jun 1793 settlement) - faster than Puerto Rico (~6 yrs), Greece (8) or Argentina (15).

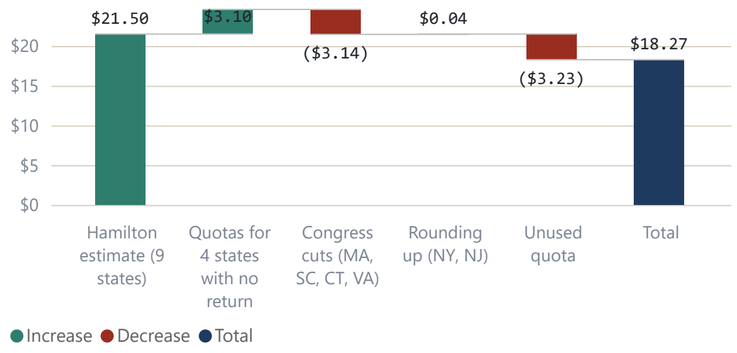
Basis of preparation

Five primary ledgers, each footed before use (Schedule B). One does not foot: two line errors in the 1792 subscription table ($500,000 and $30,000), reconciled and disclosed.

1790 specie dollars; 2025 dollars at 36.3x CPI. $21.5M was ~11% of 1790 GDP (a modern reconstruction), about $3.4 trillion in today's economy.

Bridge: authorized to assumed, 1790 $ millions

2025 $: $780M authorized -> $663M assumed.



Roll-forward schedule

Foots to $18,271,786 assumed (Bayley).

Roll-forward	1790 $	2025 $
Hamilton estimate (9 states)	$21,501,206	$780,493,
Quotas for 4 states with no return	$3,100,000	$112,530,
Congress cuts (MA, SC, CT, VA)	($3,144,950)	($114,161,(
Rounding up (NY, NJ)	$43,744	$1,587,
Unused quota	($3,228,214)	($117,184,1
Total	**$18,271,786**	**$663,265,**

Summary ledger by state, 1790 $ (net position also in 2025 $)

Estimate = Schedule E, Jan 1790. Quota = Funding Act. Assumed = Treasury final. Settlement = Commissioners, 1793. Net = assumed + settlement. Blank = no return filed.

State	Hamilton estimate	Act quota	Assumed (final)	1793 settlement	Net position	Net position (2025 $)
Massachusetts	$5,226,801	$4,000,000	$3,981,733	$1,248,801	$5,230,534	$189,868,386
South Carolina	$5,386,232	$4,000,000	$3,999,652	$1,205,978	$5,205,630	$188,964,359
Virginia	$3,680,743	$3,500,000	$2,934,416	($100,879)	$2,833,537	$102,857,393
Connecticut	$1,951,173	$1,600,000	$1,600,000	$619,121	$2,219,121	$80,554,092
North Carolina		$2,400,000	$1,793,804	($501,082)	$1,292,722	$46,925,803
New Jersey	$788,681	$800,000	$695,203	$49,030	$744,233	$27,015,647
Pennsylvania	$2,200,000	$2,200,000	$777,983	($76,709)	$701,274	$25,456,264
Rhode Island		$200,000	$200,000	$299,611	$499,611	$18,135,879
Total	**$21,501,206**	**$21,500,000**	**$18,271,786**	**$0**	**$18,271,786**	**$663,265,849**

B

Tie-out: three documents foot to the penny, one to Hamilton's own rounding, one has three cell errors in the published copy.

Each table was footed against its own printed total before any analysis. Ties = variance under $1. Ties (rounded) = Schedule E, $1,206 against Hamilton's 'about twenty-one millions and a half'. Immaterial = under $100. Does not foot = investigated and disclosed below.

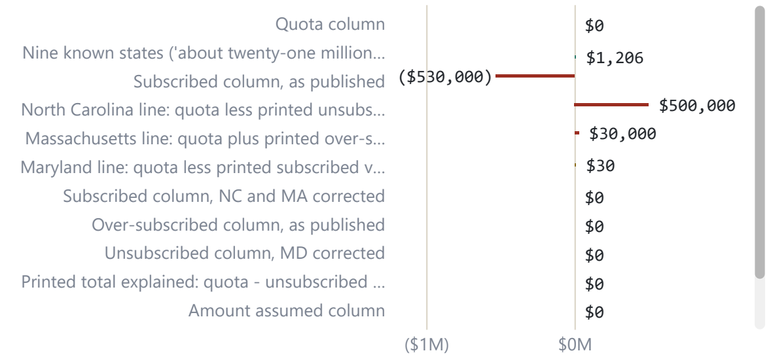
Footing schedule

Variance = computed less stated. Negatives in parentheses.

Document	Line footed	Stated	Computed	Variance	Status
Funding Act sec. 14, Aug 1790	Quota column	$21,500,000	$21,500,000	$0	Ties
Schedule E, Jan 1790	Nine known states ('about twenty-one millions and a half')	$21,500,000	$21,501,206	$1,206	Ties (rounde
Enclosure D, Jan 1792	Subscribed column, as published	$18,328,186	$17,798,186	($530,000)	Does not foo
Enclosure D, Jan 1792	North Carolina line: quota less printed unsubscribed vs printed subscribed	$1,166,356	$1,666,356	$500,000	Does not foo
Enclosure D, Jan 1792	Massachusetts line: quota plus printed over-subscribed vs printed subscribed	$4,447,014	$4,477,014	$30,000	Does not foo
Enclosure D, Jan 1792	Maryland line: quota less printed subscribed vs printed unsubscribed	$500,745	$500,775	$30	Immaterial
Enclosure D, Jan 1792	Subscribed column, NC and MA corrected	$18,328,186	$18,328,186	$0	Ties
Enclosure D, Jan 1792	Over-subscribed column, as published	$1,255,852	$1,255,852	$0	Ties
Enclosure D, Jan 1792	Unsubscribed column, MD corrected	$4,427,666	$4,427,666	$0	Ties

Variance by line footed, 1790 $

Red = does not foot.



Enclosure D line-level tie-out: quota less subscribed = unsubscribed

Statement of Subscriptions to the Loan, Treasury Department, 25 January 1792. Totals row reproduces the document's footing problem.

State	Quota	Subscribed (as printed)	Unsubscribed (as printed)	Unsubscribed (computed)	Variance	Status	Subscribed (reconciled)
New Hampshire	$300,000	$242,501	$57,499	$57,499	$0	Ties	$242,501
Massachusetts	$4,000,000	$4,447,014	$0	$0	($30,000)	Does not foot	$4,477,014
Rhode Island	$200,000	$344,259	$0	$0	$0	Ties	$344,259
Connecticut	$1,600,000	$1,455,332	$144,668	$144,668	($0)	Ties	$1,455,332
New York	$1,200,000	$1,028,239	$171,761	$171,761	$0	Ties	$1,028,239
New Jersey	$800,000	$599,704	$200,296	$200,296	($0)	Ties	$599,704
Pennsylvania	$2,200,000	$675,101	$1,524,899	$1,524,899	$0	Ties	$675,101
Delaware	$200,000	$53,306	$146,694	$146,694	$0	Ties	$53,306
Maryland	$800,000	$299,225	$500,745	$500,775	$30	Immaterial	$299,225
Virginia	$3,500,000	$2,552,571	$947,429	$947,429	$0	Ties	$2,552,571
North Carolina	$2,400,000	$1,166,356	$733,644	$1,233,644	$500,000	Does not foot	$1,666,356
South Carolina	$4,000,000	$4,634,579	$0	$0	($0)	Ties	$4,634,579
Georgia	$300,000	$300,000	$0	$0	$0	Ties	$300,000
Total	**$21,500,000**	**$17,798,186**	**$4,427,636**	**$4,927,666**	**$470,030**		**$18,328,186**

Tie-out procedure

1. Foot every column of every source table and compare to the total the document prints.

2. Where a total does not foot, tie each line: quota less subscribed must equal unsubscribed.

3. Isolate the line, quantify the variance, decide materiality, disclose.

Finding

Enclosure D (Treasury, 25 Jan 1792), as published on Founders Online, prints subscriptions totalling $18,328,186.21. The column sums to $17,798,186.21. Line-level tie-out isolates $500,000 in North Carolina: quota $2,400,000 less printed unsubscribed $733,644.43 gives $1,666,355.57 subscribed, but the document prints $1,166,355.57.

The remaining $30,000 is Massachusetts: quota $4,000,000 plus the printed over-subscription $477,013.81 gives $4,477,013.81 subscribed, but the document prints $4,447,013.81. Maryland's unsubscribed line is $30 out. With those three cells corrected, every column foots to the printed total to the penny.

The Treasury's arithmetic was right; the published cells are wrong. Four printings compared (1792 pamphlet, Seybert 1818, American State Papers 1832, Founders Online): the errors enter with the 1832 reprint. Every other document ties.

C

The ledger: each state carried through estimate, quota, subscription, assumption and settlement.

Columns are documents. Schedule E (Jan 1790), Funding Act sec. 14 (Aug 1790), Enclosure D (Jan 1792), Treasury final (Bayley, to 1793), Commissioners' settlement (Jun 1793). Blank = no return filed.

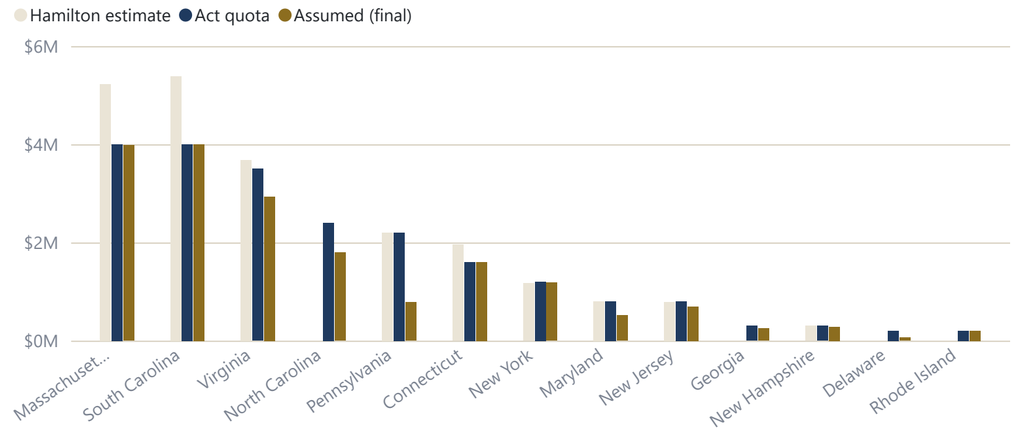
General ledger: state x document, 1790 $

Read across a row to follow one state. Totals row foots each column to its Schedule B figure.

State	1. Estimate 1790	2. Quota 1790	3. Subscribed 1792	4. Assumed (final)	5. Settlement 1793	6. Net (4 + 5)
Connecticut	$1,951,173	$1,600,000	$1,455,332	$1,600,000	$619,121	$2,219,121
Delaware		$200,000	$53,306	$59,162	($612,428)	($553,266)
Georgia		$300,000	$300,000	$246,031	$19,988	$266,019
Maryland	$800,000	$800,000	$299,225	$517,491	($151,640)	$365,851
Massachusetts	$5,226,801	$4,000,000	$4,477,014	$3,981,733	$1,248,801	$5,230,534
New Hampshire	$300,000	$300,000	$242,501	$282,596	$75,055	$357,651
New Jersey	$788,681	$800,000	$599,704	$695,203	$49,030	$744,233
New York	$1,167,575	$1,200,000	$1,028,239	$1,183,717	($2,074,846)	($891,129)
North Carolina		$2,400,000	$1,666,356	$1,793,804	($501,082)	$1,292,722
Pennsylvania	$2,200,000	$2,200,000	$675,101	$777,983	($76,709)	$701,274
Rhode Island		$200,000	$344,259	$200,000	$299,611	$499,611
South Carolina	$5,386,232	$4,000,000	$4,634,579	$3,999,652	$1,205,978	$5,205,630
Virginia	$3,680,743	$3,500,000	$2,552,571	$2,934,416	($100,879)	$2,833,537
Total	**$21,501,206**	**$21,500,000**	**$18,328,186**	**$18,271,786**	**$0**	**$18,271,786**

Estimate, quota, assumed - by state, 1790 $

No pale bar = no return on file. MA and SC reported $5.2M / $5.4M and were capped at $4.0M.



Variance schedule: Act quota less Hamilton's estimate

Congress cut ($3,144,950) from the four largest reported debts and wrote NH, PA, MD in at exactly Hamilton's round numbers. L...

State	Hamilton estimate	Act quota	Quota less estimate	Act quota (2025 $)
South Carolina	$5,386,232	$4,000,000	($1,386,232)	$145,200,000
Massachusetts	$5,226,801	$4,000,000	($1,226,801)	$145,200,000
Connecticut	$1,951,173	$1,600,000	($351,173)	$58,080,000
Virginia	$3,680,743	$3,500,000	($180,743)	$127,050,000
Delaware		$200,000		$7,260,000
Georgia		$300,000		$10,890,000
North Carolina		$2,400,000		$87,120,000
Rhode Island		$200,000		$7,260,000
Maryland	$800,000	$800,000	$0	$29,040,000
New Hampshire	$300,000	$300,000	$0	$10,890,000
Pennsylvania	$2,200,000	$2,200,000	$0	$79,860,000
Total	**$21,501,206**	**$21,500,000**	**($3,101,206)**	**$780,450,000**

D

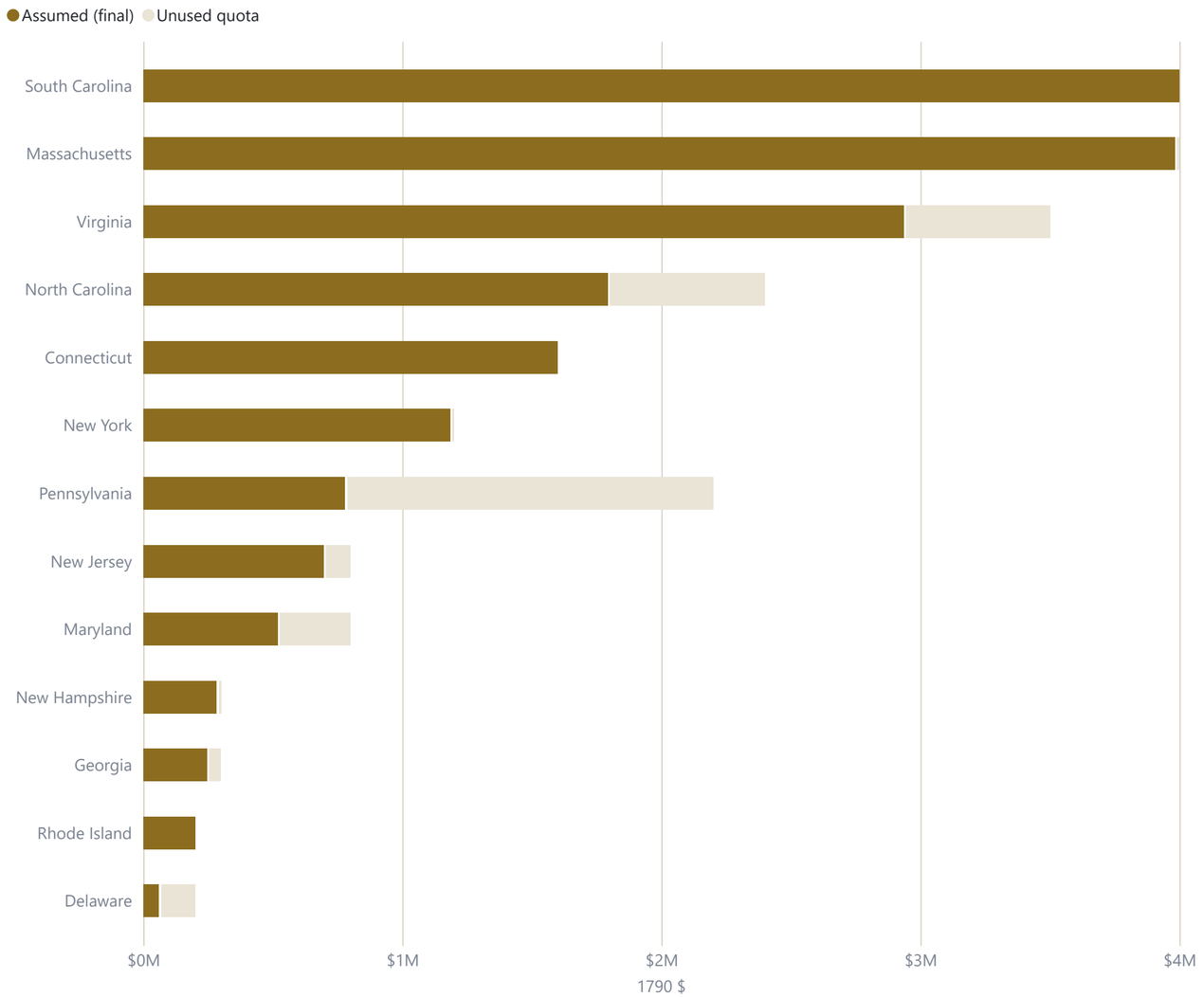
Assumption was a ceiling, not a cheque: 85% of the $21.5M was taken up.

Creditors had to bring state paper to a federal loan office and exchange it for federal stock. Utilization shows whether each quota matched real outstanding debt. Sources: Enclosure D (first window, Jan 1792); Bayley, History of the National Loans (Treasury) for final amounts.

Prepared by R. Barry
1790 $ · [2025 $ at 36.3x]

Quota used and unused, by state, 1790 $

Bar length = Act quota. Gold = assumed; pale = left on the table.



Utilization: $18.27M assumed of $21.50M authorized ($663M of $780M in 2025 $)

Needle = assumed; end of dial = authorized.



Take-up schedule by state

Subscribed = first window to Sep 1791 (MA, RI, SC subscribed above quota; the quota was a ceiling). Assumed = final after extensio…

State	Act quota	Subscribed Jan 1792	Assumed (final)	Unused quota	Take-up %	Assumed (2025
Connecticut	$1,600,000	$1,455,332	$1,600,000	$0	100.0	$58,080,0
Rhode Island	$200,000	$344,259	$200,000	$0	100.0	$7,260,0
South Carolina	$4,000,000	$4,634,579	$3,999,652	$348	100.0	$145,187,3
Massachusetts	$4,000,000	$4,477,014	$3,981,733	$18,267	99.5	$144,536,9
New York	$1,200,000	$1,028,239	$1,183,717	$16,283	98.6	$42,968,9
New Hampshire	$300,000	$242,501	$282,596	$17,404	94.2	$10,258,2
New Jersey	$800,000	$599,704	$695,203	$104,797	86.9	$25,235,8
Virginia	$3,500,000	$2,552,571	$2,934,416	$565,584	83.8	$106,519,3
Georgia	$300,000	$300,000	$246,031	$53,969	82.0	$8,930,9
North Carolina	$2,400,000	$1,666,356	$1,793,804	$606,196	74.7	$65,115,0
Maryland	$800,000	$299,225	$517,491	$282,509	64.7	$18,784,9
Pennsylvania	$2,200,000	$675,101	$777,983	$1,422,017	35.4	$28,240,8
Delaware	$200,000	$53,306	$59,162	$140,838	29.6	$2,147,5
Total	**$21,500,000**	**$18,328,186**	**$18,271,786**	**$3,228,214**	**85.0**	**$663,265,8**

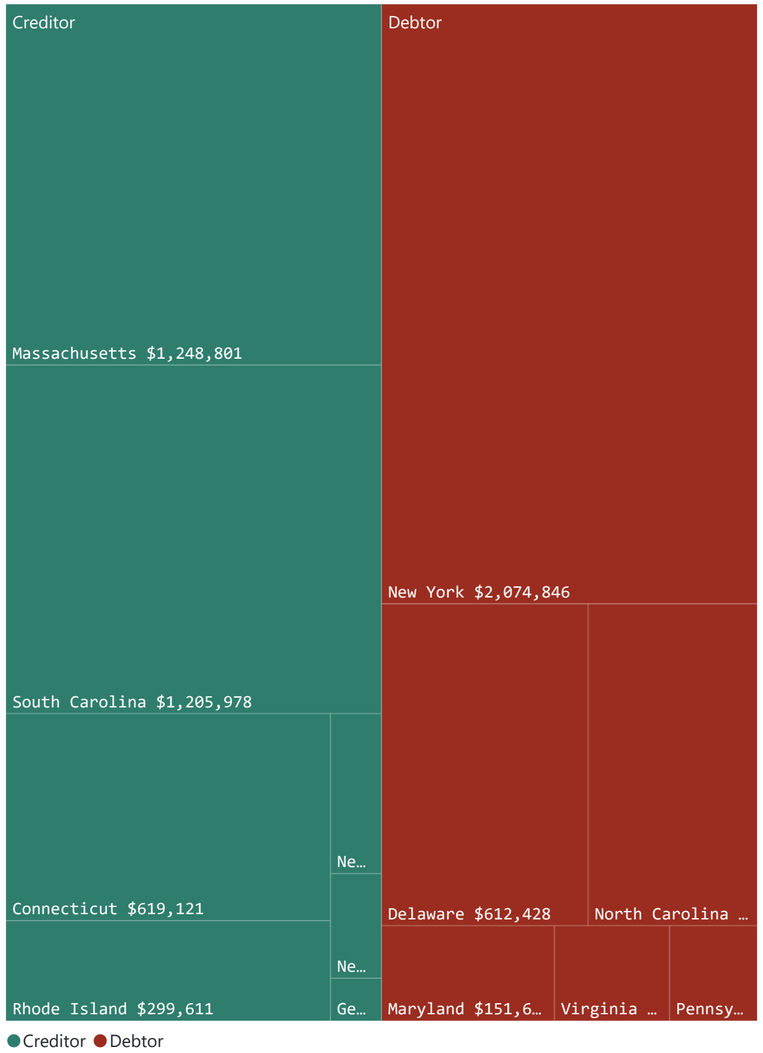
E

The true-up: the 1793 settlement of war accounts, posted as a T-account. Both sides foot to $3,517,584.

This is the audit Madison demanded before assumption. Each state's war expenditure was netted against its share of the common cost. Posted as 'the States in account with the United States': a debit balance means the state owes the Union (debtor); a credit balance means the Union owes the state (creditor). Virginia is on the debit side. Source: Commissioners to Washington, 29 Jun 1793.

The two sides of the account, to scale

Tile area = absolute balance. The teal block and the red block are the same size: $3,517,584 each...



Dr · Debtor states: balances due FROM the state

Six states under-contributed to the common cost. Shown as negatives (in parentheses) to ma...

State	Dr - due from state	1793 settlement (2025 $)	Settlement per he
New York	($2,074,846)	($75,316,910)	($6.
Delaware	($612,428)	($22,231,136)	($10.
North Carolina	($501,082)	($18,189,277)	($1.
Maryland	($151,640)	($5,504,532)	($0.
Virginia	($100,879)	($3,661,908)	($0.
Pennsylvania	($76,709)	($2,784,537)	($0.
Total	**($3,517,584)**	**($127,688,299)**	**($1.**

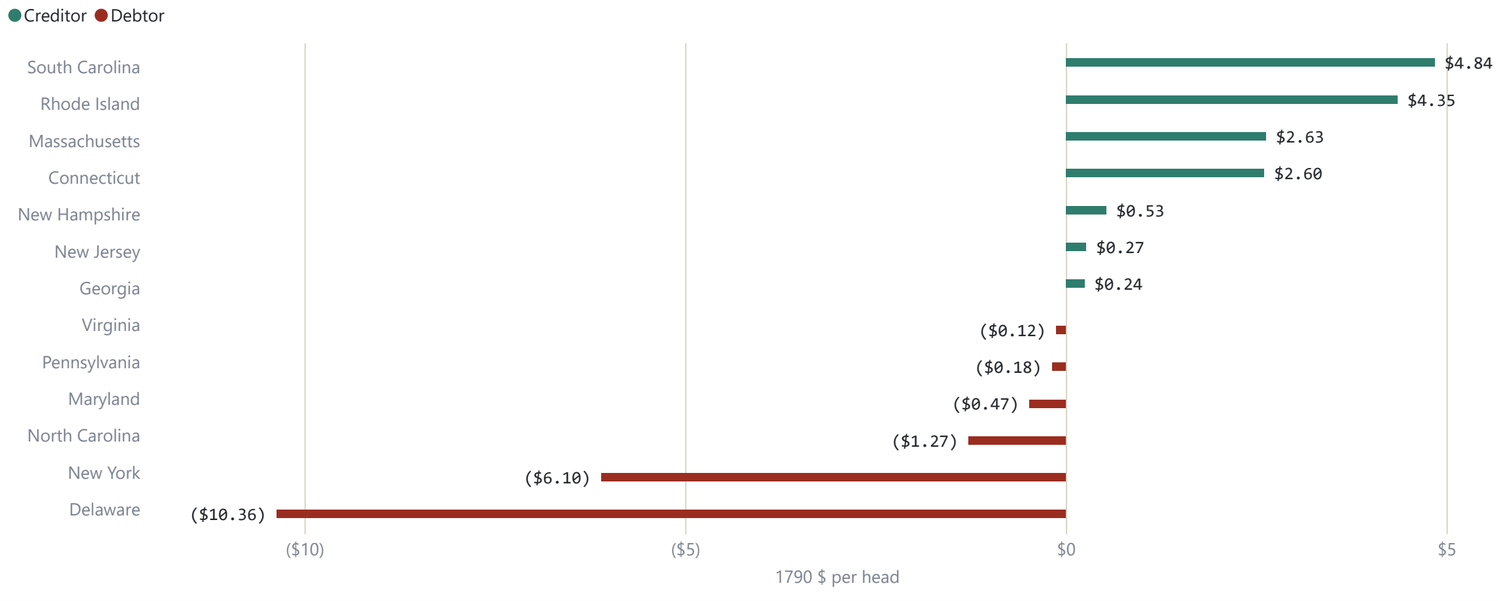
Cr · Creditor states: balances due TO the state

Seven states over-contributed to the common cost.

State	Cr - due to state	1793 settlement (2025 $)	Settlement per hea
Massachusetts	$1,248,801	$45,331,476	$2.6
South Carolina	$1,205,978	$43,777,001	$4.8
Connecticut	$619,121	$22,474,092	$2.6
Rhode Island	$299,611	$10,875,879	$4.3
New Hampshire	$75,055	$2,724,497	$0.5
New Jersey	$49,030	$1,779,789	$0.2
Georgia	$19,988	$725,564	$0.2
Total	**$3,517,584**	**$127,688,299**	**$2.4**

Settlement balance per head, 1790 $

New York ($6.10) per head [($221) today]. South Carolina $4.84 [$176]. Virginia ($0.12). Negatives in parentheses, 2025 $ in brackets.

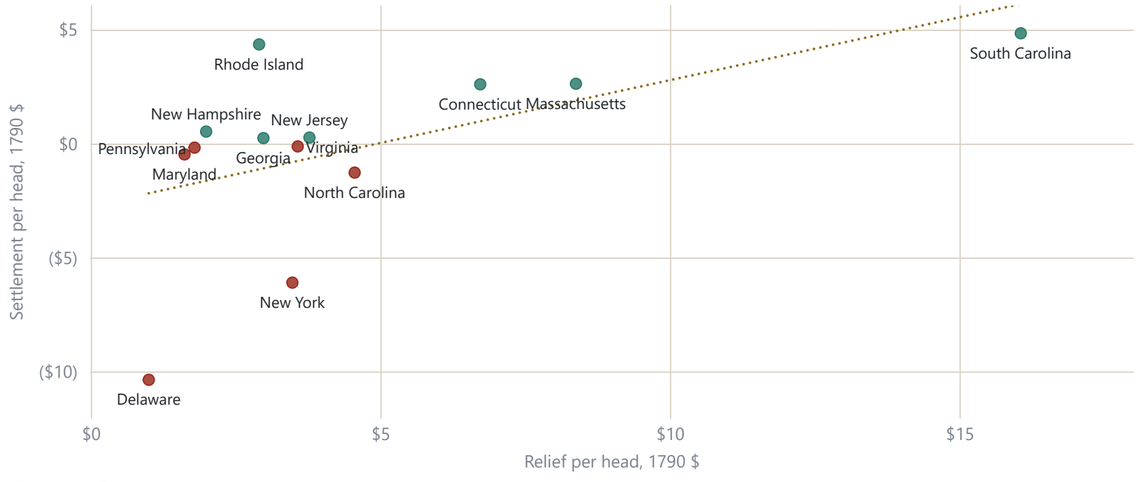


Relief tracked contribution: creditor states received 2.3x the relief per head of debtor states.

Hamilton's defence was that the state debts were incurred for a common cause, so relieving them was rough justice. If so, states the 1793 audit found had over-paid should have received more relief. They did: $6.12 vs $2.67 per head (exact permutation p = 0.047, uncorrected; Spearman rho = 0.55 per head, 0.31 on raw dollars). Suggestive, not confirmatory - see Schedule H.

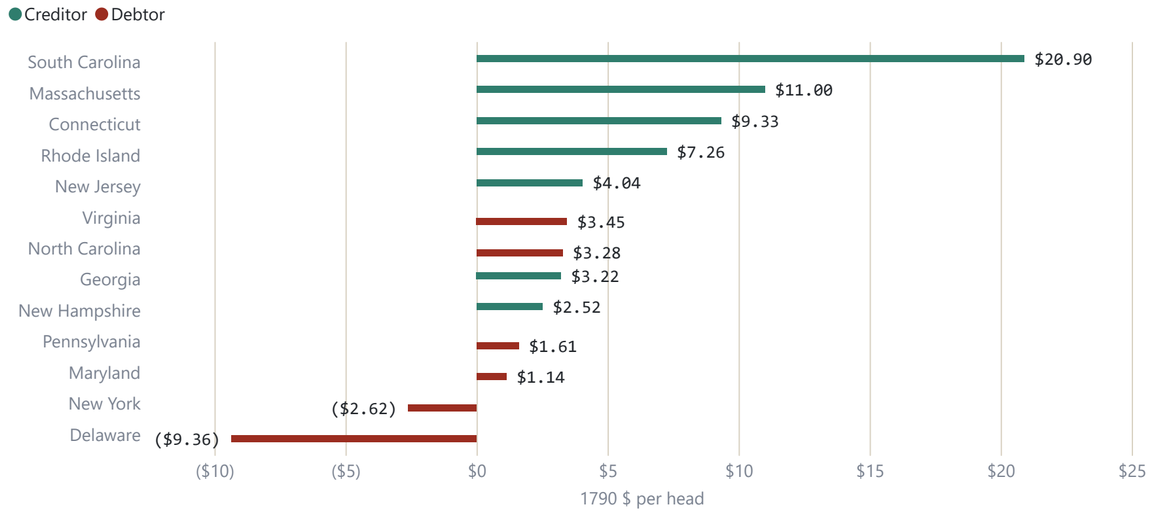
Relief received vs. 1793 settlement, per head, 1790 $

One point per state. Dotted line = linear trend.



Net position per head: relief plus settlement, 1790 $

Eleven of thirteen came out ahead. South Carolina $20.90 [$759 today]; Delaware ($9.36); New York ($2.62). Negatives in parentheses, 2025 $ in bra...



Net federal position schedule, 1790 $ and 2025 $

Relief = finally assumed. Net = relief + settlement. Totals row: relief $18,271,786; settlement nil; net $18,271,786.

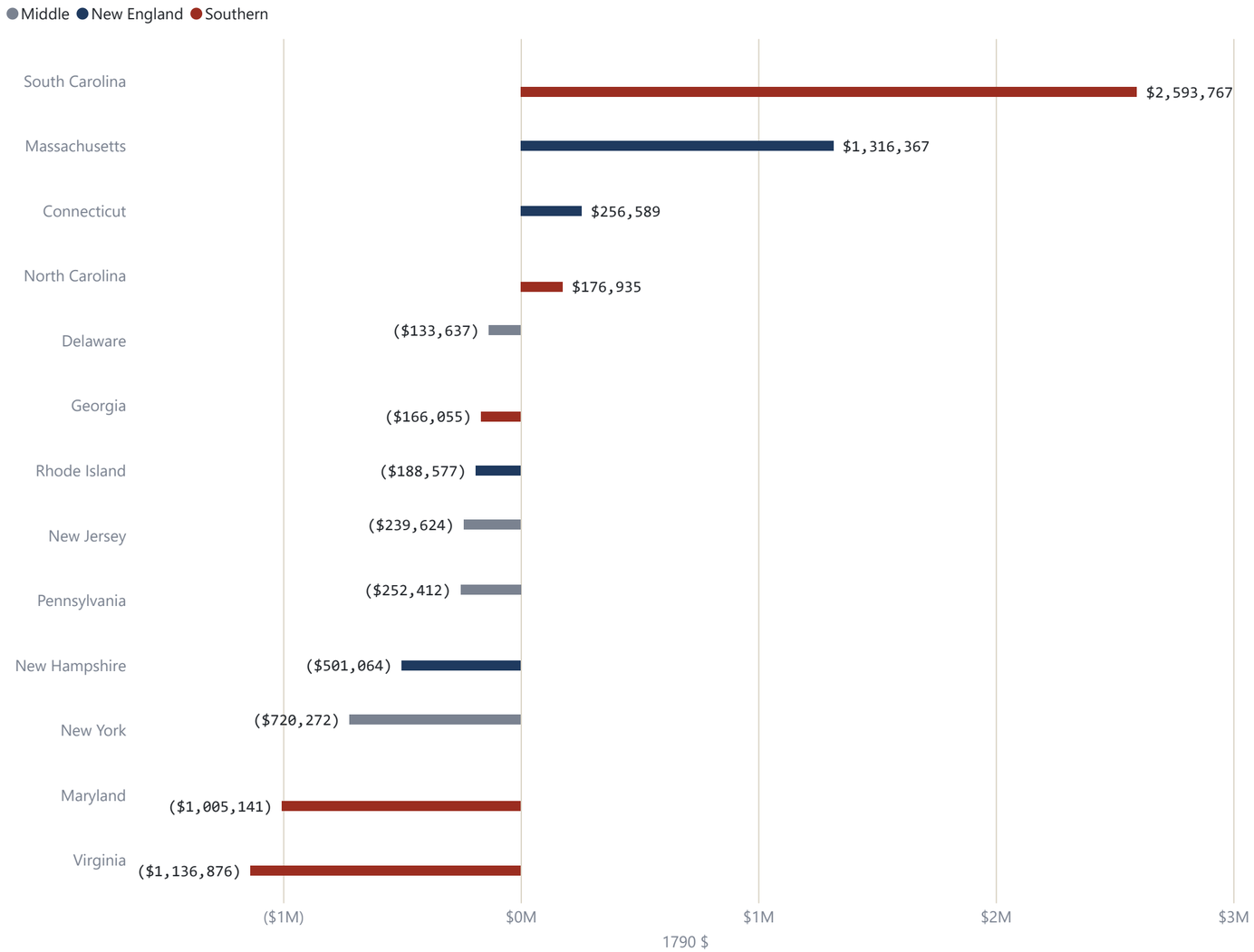
State	Region	1793 position	Assumed (final)	1793 settlement	Net position	Net position (2025 $)	Net per head	Net per head (2025 $)	Pop. share %
South Carolina	Southern	Creditor	$3,999,652	$1,205,978	$5,205,630	$188,964,359	$20.90	$758.67	6.5
Massachusetts	New England	Creditor	$3,981,733	$1,248,801	$5,230,534	$189,868,386	$11.00	$399.45	12.5
Connecticut	New England	Creditor	$1,600,000	$619,121	$2,219,121	$80,554,092	$9.33	$338.54	6.2
Rhode Island	New England	Creditor	$200,000	$299,611	$499,611	$18,135,879	$7.26	$263.51	1.8
New Jersey	Middle	Creditor	$695,203	$49,030	$744,233	$27,015,647	$4.04	$146.71	4.8
Virginia	Southern	Debtor	$2,934,416	($100,879)	$2,833,537	$102,857,393	$3.45	$125.24	21.6
North Carolina	Southern	Debtor	$1,793,804	($501,082)	$1,292,722	$46,925,803	$3.28	$119.18	10.3
Georgia	Southern	Creditor	$246,031	$19,988	$266,019	$9,656,480	$3.22	$116.98	2.2
New Hampshire	New England	Creditor	$282,596	$75,055	$357,651	$12,982,714	$2.52	$91.50	3.7
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
Total			**$18,271,786**	**$0**	**$18,271,786**	**$663,265,849**	**$4.80**	**$174.17**	**100.0**

Not proportional to population: one dollar in five sits in a different state than a per-head split would put it.

What each state would have received had $21.5M been split like congressional apportionment, versus the Act. Choose the population basis at right; the answer barely moves. No detectable 'North vs South' tilt (permutation p = 0.78, low power at n = 4 vs 5): South Carolina alone drives the Southern mean.

Quota less population-proportional share, 1790 $

Right of zero = more than its head-count share. Negatives in parentheses.



Population basis (pick one)

- ☐ Free population
- ☐ Three-fifths basis
- ■ Total population

Apportionment schedule

Gaps net to zero by construction. Dissimilarity: total population 20.2%, free population 19.1%, three-fifths basis 19.8%.

State	Act quota	Per-head share	Quota less per-head share	Gap (2025 $)
South Carolina	$4,000,000	$1,406,233	$2,593,767	$94,153,749
Massachusetts	$4,000,000	$2,683,633	$1,316,367	$47,784,137
Connecticut	$1,600,000	$1,343,411	$256,589	$9,314,172
North Carolina	$2,400,000	$2,223,065	$176,935	$6,422,726
Delaware	$200,000	$333,637	($133,637)	($4,851,016)
Georgia	$300,000	$466,055	($166,055)	($6,027,795)
Rhode Island	$200,000	$388,577	($188,577)	($6,845,335)
New Jersey	$800,000	$1,039,624	($239,624)	($8,698,356)
Pennsylvania	$2,200,000	$2,452,412	($252,412)	($9,162,548)
New Hampshire	$300,000	$801,064	($501,064)	($18,188,613)
New York	$1,200,000	$1,920,272	($720,272)	($26,145,872)
Maryland	$800,000	$1,805,141	($1,005,141)	($36,486,635)
Virginia	$3,500,000	$4,636,876	($1,136,876)	($41,268,614)
Total	**$21,500,000**	**$21,500,000**	**$0**	**$0**

Test schedule

A: regional tilt in quotas? B: did relief track 1793 contribution? C: proportional to population? p-values are uncorrected for multiple comparisons; C is descriptive (no p-value).

Question	Test	Statistic	p-value	Reading
Regional tilt	Welch t-test, NE vs South quota per head	t = -0.51	0.627	NE $5.04 vs South $6.51 per head. No
Regional tilt	Mann-Whitney U, NE vs South	U = 9	0.905	Rank-based version of the same compar
Regional tilt	Exact permutation, NE vs South (126 relabelings)	diff = ($1.47)	0.778	Every possible relabeling of 9 states
Relief vs contribution	Spearman rho, relief per head vs 1793 settlement per head	rho = 0.55	0.052	Moderate positive rank correlation: s
Relief vs contribution	Pearson r, same variables	r = 0.54	0.056	Linear version; agrees in size.
Relief vs contribution	Spearman rho on raw dollars (no shared population denominator)	rho = 0.31	0.306	Same sign, weaker. Per-capita ratios
Relief vs contribution	Exact permutation, creditor vs debtor states, one-sided (1716 relabelings)	$6.12 vs $2.67	0.047	Creditor states received 2.3x the rel

Sources

Schedule E, Report Relative to a Provision for the Support of Public Credit, 9 Jan 1790 (Founders Online).

An Act making provision for the Debt of the United States, 4 Aug 1790, 1 Stat. 138, sec. 14.

Enclosure D, Statement of Subscriptions to the Loan, Treasury Department, 25 Jan 1792 (Founders Online).

Bayley, History of the National Loans of the United States (U.S. Treasury, 1881), p. 33.

Commissioners for Settling Accounts to George Washington, 29 Jun 1793 (Founders Online).

Return of the Whole Number of Persons (First Census, 1790), U.S. Census Bureau.

How it was built

Python (pandas, scipy) reads the five source tables, foots them, joins them to the census, runs the tests and writes a ledger CSV. This report is a Power BI project (.pbip): the semantic model and every page and visual are generated by src/build_model.py and src/build_pbir.py, so the dashboard is version-controlled like code.

github.com/josephrbarry/hamilton-debt-assumption